Loading the battery dataset

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%% STEP 1: Download and Load Battery Dataset
dataFile
= matlab.internal.examples.downloadSupportFile("predmaint","batteryagingdata/
singlecell/v1/singleCellLifeTimeData.zip");
unzip(dataFile);
load("singleCellLifeTimeData.mat");

%% STEP 2: Parse and Segment Data
bParser = batteryTestDataParser(data);
bParser.CurrentVariable = 'Current';
bParser.VoltageVariable = 'Voltage';
bParser.TimeVariable = 'DateTime';
bParser.CycleIndexVariable = 'Cycle_Index';
bParser.StepIndexVariable = 'Step_Index';
segmentedRawDataTable = segmentData(bParser);

%% STEP 3: Extract All Valid "Charge" Data for Cycle 1
chargingData = segmentedRawDataTable(...
    segmentedRawDataTable.Cycle_Index == 1 & ...
    segmentedRawDataTable.CyclingPhases == "Charge", :);
chargingData.TimeInSeconds = seconds(chargingData.DateTime -
chargingData.DateTime(1));
% 1. Calculate time in seconds starting from 0
t = chargingData.TimeInSeconds;

% 2. Calculate instantaneous power loss at each measurement point (P = I^2 * R)
P_loss = (chargingData.Current .^ 2) .* chargingData.Internal_Resistance;

% 3. Integrate power over time to get total energy loss in Joules (J)
energyLossJoules = trapz(t, P_loss);

% 4. Convert Joules to Watt-hours (1 Wh = 3600 Joules)
energyLossWh = energyLossJoules / 3600;

% 5. --- Display Results ---
fprintf('--- Cycle 1 Resistive Loss (Charge Phase) ---\n');
```

```
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fprintf('Resistive Energy Loss: %.4f Joules (J)\n', energyLossJoules);
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Resistive Energy Loss: 335.5903 Joules (J)
```

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fprintf('Resistive Energy Loss: %.6f Watt-hours (Wh)\n', energyLossWh);
```

```
Resistive Energy Loss: 0.093220 Watt-hours (Wh)
```